

The separate universe computes a different variable: an exact criterion for $\delta N = \zeta$ in non-attractor phases

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(Dated: September 4, 2026)

The isotropic separate universe (δN) computes δN_c , the e-fold count of a fluid worldline labelled by its position on the initial slice; the standard δN formula identifies δN_c with Maldacena's ζ . We show this identification is an exact, invertible change of variable, $\delta N_c = \zeta_{L,f}[1 - I/3] + O(k_L^2/a^2 H^2)$, with $I = \zeta_{L,f}^{-1} \int (\epsilon/c_s^2) \dot{\zeta}_L dt$ the ζ -growth-weighted mean: $\delta N_c = \zeta$ at linear order iff $I \rightarrow 0$. At second order the worldline label matters: the initial-position label (the separate universe's own label) reproduces the in-in monopole exactly, -5 , for every constant ϵ ; the final-position label differs by a translation term with zero monopole. We validate on four backgrounds – attractor slow roll and ekpyrosis ($I = 0$, identity map), ultra-slow-roll ($I = O(\epsilon)$, small error), and matter-dominated contraction ($I \rightarrow \epsilon$, an $O(1)$ identification error) – and place the criterion against the non-attractor consistency-relation and beyond- δN gradient-expansion literature. There is no discrepancy in the physics, only in which variable δN_c is compared to.

I. MOTIVATION

The squeezed limit of the primordial bispectrum is governed, in single-clock inflation, by Maldacena's consistency relation, and can equivalently be computed by the separate-universe (δN) formalism, which treats the long-wavelength mode as a local shift of the background evolution [1]. In a non-attractor phase, where $\dot{\zeta} \neq 0$ on super-Hubble scales, two independent difficulties arise. First, Namjoo, Firouzjahi & Sasaki showed that ultra-slow-roll (USR) inflation violates the consistency relation, $f_{\text{NL}} = 5/2$, and that δN built from $N(\phi)$ alone misses this: the growing mode is an independent initial datum, requiring $N(\phi, \pi)$ to be correctly captured [3]. This initial-data requirement was extended to $P(X)$ non-attractors by Chen, Firouzjahi, Namjoo & Sasaki, who found $f_{\text{NL}} = 5(1 + c_s^2)/(4c_s^2)$ and confirmed δN with momentum agrees [4]. Pajer, Schmidt & Zaldarriaga separately argued that in genuine attractors the squeezed bispectrum is an unobservable coordinate artifact of the long mode [5], a statement later sharpened with conformal Fermi coordinates by Dai, Pajer & Schmidt [6]. Cai *et al.* tracked f_{NL} through smooth and sharp USR transitions and found δN and in-in agree throughout [7]. Bravo, Mooij, Palma & Pradenas derived a generalized consistency relation for non-attractor backgrounds and showed the canonical single-field local f_{NL} is removable by a change of physical coordinates [8, 9], and Passaglia, Hu & Motohashi confirmed $\delta N(\phi, \pi)$ /in-in agreement for PBH-motivated USR scenarios [10]. Separately, Artigas, Grain & Vennin, and Jackson *et al.*, identified a phase-space separate-universe breakdown localized to a finite window of scales at slow-roll-to-USR transitions, traced to a momentum/initial-data mismatch [11, 12].

Second, and logically independent, a companion theory-audit note on this program found that

the isotropic separate universe applied to a matter-dominated cosmological contraction ($w = 0$, $\epsilon = 3/2$) returns $f_{\delta N} = -5$ (initial-position label) while the from-scratch in-in squeezed bispectrum of ζ is $-35/16 + (15/16)\mu^2$ [21], a monopole gap of $25/8$ – an $O(1)$ discrepancy, not a small correction. Cai, Xue, Brandenberger & Zhang quote a squeezed amplitude of $-35/8$ for the same background [18]; the from-scratch value is uniformly a factor of 2 smaller, and [21] locates the slip in their amplitude step rather than in any vertex or commutator, reproducing their shape function monomial by monomial. Were $-35/8$ the correct in-in monopole instead of $-15/8$, the gap to $f_{\delta N}^{\text{init}} = -5$ would be $-5/8$, a factor of $8/7$ rather than $8/3$; the headline $O(1)$ statement of this note rests on the from-scratch value. A second companion note derived the exact second-order map between the two variables and showed the gap decomposes into a linear rescaling and an explicit second-order term, both proportional to ϵ [22]. Every prior test of δN /in-in agreement in the literature above is an inflationary non-attractor, where $\epsilon \ll 1$ throughout the growth of the long mode. This note asks the question directly: is there a single criterion, applicable to both inflationary and contracting non-attractors, that predicts when the two constructions agree and by how much they differ when they do not.

II. THE THREADING IDENTITY AND THE MAP

Comoving gauge, ADM variables $N = 1 + \alpha$, $N_i = \partial_i \psi + \tilde{N}_i$. The separate universe evolves each patch as a homogeneous FRW region identified by its fluid element's position x_i on a flat, super-Hubble initial slice Σ_i ($k_L \ll a_i H_i$); its output is the e-fold count δN_c of the fluid (normal) congruence, labelled by x_i , from Σ_i to the comoving final slice Σ_f . Along a fluid worldline $\dot{x}^i = -N^i$ the fully nonlinear relation between the local expansion

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and the volume element gives the exact identity

$$\delta N_c(x_i) = \zeta(t_f, x_f) - \frac{1}{3} \int_{t_i}^{t_f} \partial_i N^i(t, x(t)) dt, \quad x_f = x(t_f), \quad (1)$$

with $\delta \rho_c / \rho \propto \partial_i N^i$ up to a gradient term. On super-Hubble scales, $\partial_i N^i = (\epsilon / c_s^2) \dot{\zeta} - \nabla^2 \zeta / (a^2 H)$ for the $P(X)$ class (a genuine fluid with non-adiabatic pressure adds a term) [1, 2]. Splitting the interval at the comoving slice through x_i and integrating,

$$\delta N_c = \zeta_{L,f} \left[1 - \frac{I}{3} \right] + O\left(\frac{k_L^2}{a^2 H^2}\right), \quad I \equiv \frac{1}{\zeta_{L,f}} \int_{t_i}^{t_f} \frac{\epsilon}{c_s^2} \dot{\zeta} dt, \quad (2)$$

exact on super-Hubble scales for any single-field history from a flat, super-Hubble initial slice; the dropped gradient term is $O(k_L^2 / a_i H_i^2)$ or $O(k_L^2 / a_f H_f^2)$, whichever end of the interval dominates. $I \rightarrow 0$ (identity map) on any constant mode, including the attractor and the dominant ekpyrotic mode; $I = 1 - \lambda$ reduces to $\epsilon/3$ for constant ϵ , so $\lambda \equiv 1 - I/3 = 1 - \epsilon/3$, and to $\lambda_{\text{USR}} = 1 + \epsilon_f/3 - \sqrt{\epsilon_s \epsilon_f}/3$ for USR with $\epsilon = \epsilon_s(a/a_s)^{-6}$ from a slow-roll-frozen $\zeta_{L,i} \neq 0$ at a_s , $\zeta_L \propto a^3$ thereafter. At second order, for constant ϵ and $c_s = 1$, solving the exact ADM constraints to $O(\zeta_L \zeta_S)$ and integrating Eq. (1) gives $f_{\delta N} = f^{\text{in-in}}/\lambda + f_{\text{map}}$, where f_{map} depends on how the fluid worldline is labelled. Labelled by *initial* position x_i – the label the separate universe uses, since it never learns where the fluid element ends up –

$$f_{\text{map}}^{\text{init}}(\epsilon, \mu) = \frac{5\epsilon}{4(3-\epsilon)} [(\epsilon - 2) - \epsilon \mu^2], \quad (3)$$

and $f_{\delta N}^{\text{init}} \equiv f^{\text{in-in}}/\lambda + f_{\text{map}}^{\text{init}} = -5$, isotropic, for every constant ϵ . Labelled instead by *final* position $x_f = x(t_f)$, the two labels differ by a pure translation $\xi^i = -\int N_L^i dt$ along the long mode, adding a shear-plus-dilation term with zero monopole:

$$f_{\text{map}}^{\text{fin}}(\epsilon, \mu) = -\frac{5\epsilon}{4}(1 - \mu^2) = f_{\text{map}}^{\text{init}} + \frac{5\epsilon}{4(3-\epsilon)}(1 - 3\mu^2), \quad (4)$$

giving $f_{\delta N}^{\text{fin}} = -\frac{15(\epsilon-4)}{4(\epsilon-3)} + \frac{15\epsilon}{4(3-\epsilon)}\mu^2$ ($-\frac{25}{4} + \frac{15}{4}\mu^2$ at $\epsilon = 3/2$); $\mu = \hat{k}_L \cdot \hat{k}_S$. Both labels give the same, label-independent monopole $-5\epsilon/6$; only the quadrupole is a labelling convention.¹ In terms of a constant equation of state w ($\epsilon = \frac{3}{2}(1+w)$, $c_s = 1$),

$$\lambda = \frac{1-w}{2}, \quad f_{\text{map}}^{\text{fin}} = -\frac{15}{8}(1+w)(1-\mu^2), \quad (5)$$

with label-independent monopole $f_{\text{map}}^{\text{mono}} = -\frac{5}{4}(1+w)$. Fig. 1 shows both across $-1 \leq w \leq 1$: at $w = -1$ ($\epsilon = 0$,

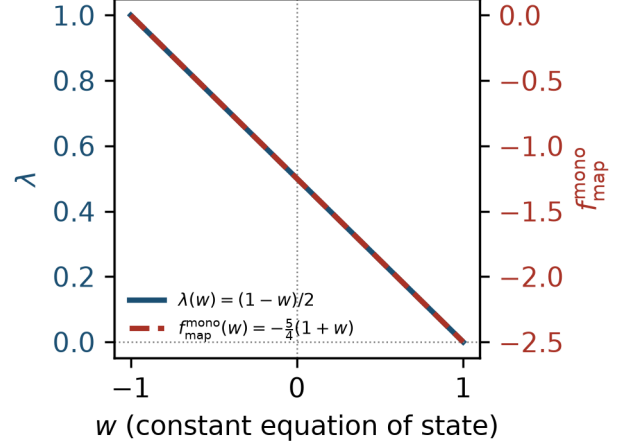


FIG. 1. The linear rescaling $\lambda(w) = (1-w)/2$ (dimensionless, left axis) and the label-independent second-order map monopole $f_{\text{map}}^{\text{mono}}(w) = -\frac{5}{4}(1+w)$ (dimensionless, right axis), Eq. (5), for constant equation of state $-1 \leq w \leq 1$ at $c_s = 1$. At $w = -1$ (attractor limit, $\epsilon = 0$) $f_{\text{map}}^{\text{mono}}$ vanishes and $\lambda \rightarrow 1$, so the map is the identity; λ vanishes at $w = 1$ (kination), where δN_c loses linear response.

de Sitter/attractor limit) $\lambda \rightarrow 1$ and $f_{\text{map}}^{\text{mono}} \rightarrow 0$, so the map is the identity; $\lambda \rightarrow 0$ at $w = 1$ (kination, $\epsilon = 3$), where δN_c ceases to be a usable linear-response variable and the general- ϵ in-in monopole $-\frac{5}{18}(\epsilon-3)(\epsilon-6)$ itself vanishes.

III. THE CRITERION

Equations (2)–(4) give a single statement:

The map $\delta N_c = \zeta_{L,f} [1 - I/3] + O(k_L^2/a^2 H^2)$, $I = \zeta_{L,f}^{-1} \int (\epsilon/c_s^2) \dot{\zeta} dt$, is exact and invertible for any single-field history ($\lambda \equiv 1 - I/3 \neq 0$); it is not a discrepancy in the physics, only a statement about which variable δN_c is. The isotropic separate universe (δN , with $N(\phi, \pi)$) computes δN_c , and the standard δN formula identifies δN_c with ζ . That identification is exact iff $I \rightarrow 0$; it carries an $O(1)$ error in the identification – not in either variable separately – whenever $I = O(1)$, e.g. every constant- ϵ contraction with a growing ζ . When $I \ll 1$ (attractor and dominant ekpyrotic mode: $\dot{\zeta}_L = 0$, $I = 0$ exactly; USR: $\epsilon \ll 1$ throughout the growth) the identification holds to $O(I)$.

Via the Hamiltonian constraint on comoving slices, $I = O(1)$ is equivalent to the long mode's Bardeen potential growing as $\Psi_L = O((aH/k)^2) \zeta_L$, the textbook signature of a non-attractor contraction [15, 16], and $I \rightarrow 0$ recovers the gradient-expansion assumption of Lyth, Malik & Sasaki [13], following Salopek & Bond [14]. Two logically distinct requirements are thus needed for the standard δN formula to give ζ in a non-attractor phase: (i) $N(\phi, \pi)$, to capture the growing mode as an indepen-

¹ Each of the five geometric contributions to $f_{\text{map}}^{\text{fin}}$ carries an overall factor of ϵ : the lapse, the second-order metric perturbation, the gradient term, the final-position worldline displacement, and the initial-label translation [22].

dent initial datum [3]; (ii) $I \rightarrow 0$, so that $\delta N_c = \zeta$. USR satisfies (ii) approximately because $\epsilon \ll 1$ while ζ grows – not because the comoving shift is small, which it is not: the shift itself is $O(1/k_L)$ in every non-attractor, USR included, but its coefficient $\epsilon \dot{\zeta}_L$ is what controls I , and this coefficient is small in USR precisely when its growing-mode initial-data problem (i) is largest. Every inflationary test cited in Sec. I checked (i) with (ii) automatically small; none tested a regime with $I = O(1)$.

Table I summarizes the four validations by exact sympy (script and manifest in the reproducibility statement). The dust case ($w = 0$) is the one already reported in Ref. [21]: the isotropic separate universe gives $f_{\delta N}^{\text{init}} = -5$ for every constant $\epsilon \in (1, 3)$, while the in-in monopole is $f_{\text{mono}}^{\text{in-in}} = -\frac{5}{18}(\epsilon - 3)(\epsilon - 6) = -15/8$ at $\epsilon = 3/2$, so the identification error, $-5 - (-15/8) = -25/8$, is a factor $8/3$; the two coincide only in the limit $\epsilon \rightarrow 0$. USR agrees because $\epsilon \ll 1$ throughout the growth of the long mode – its row tests the size of the $O(\epsilon)$ correction to the map, not a constraint from Namjoo-Firouzjahi-Sasaki’s leading-order $f_{\text{NL}} = 5/2$ itself [3]. Attractor slow roll is the identity map, consistent with Maldacena’s consistency relation [1]. Ekpyrotic contraction ($\epsilon \gg 3$) has $I = 0$ because the physical long mode sits on the constant- ζ solution, reproducing the attractor behavior of single-field ekpyrosis found by Creminelli, Nicolis & Zaldarriaga [17]; with $\dot{\zeta}_L = 0$ the row is a consistency check on the definition of I rather than a nontrivial test of the criterion. The non-dominant, growing- Ψ ekpyrotic mode would have $I = O(1)$ by the same criterion, but never sets $\zeta_L(t_f)$ and so is not physically realized. λ_{USR} assumes ζ_L is purely the growing a^3 mode from a slow-roll-frozen value at a_s ; a slow-roll-to-USR transition with a non-negligible decaying piece is not covered.

IV. WHAT IS NEW

Relative to Takamizu, Mukohyama, Sasaki & Tanaka and Naruko, Takamizu & Sasaki, who computed the $O(\epsilon^2)$ (gradient-expansion, $m = 2$) correction beyond the leading-order δN /separate-universe approximation for the superhorizon non-Gaussianity [19, 20], the term this note restores, $-\nabla^2 \zeta / (a^2 H)$ in $\partial_i N^i$, is the same known next-to-leading-order gradient-expansion object; the point here is that in a non-attractor contraction the *other*, retained term, $(\epsilon/c_s^2)\dot{\zeta}$, is already $O(k^0)$ and $O(1)$ in ϵ , so the leading-order ($m = 0$) identification $\delta N_c = \zeta$ can fail at $O(1)$ without needing the $m = 2$ correction at all; the two effects are independent and this note isolates the first one with a single closed-form criterion I . Relative to Refs. [3, 4], this note identifies a second, independent source of identification error between δN_c and ζ in non-attractors – controlled by I rather than by initial data – and shows it is $O(\epsilon)$ in USR (hence invisible to their test) and $O(1)$ in a matter-like contraction. Relative to Ref. [5], the attractor case reduces to $\Theta \equiv \epsilon \dot{\zeta}_L / (H \zeta_L) = 0$ and the identity map, so the

two curvature variables coincide there and the only remaining question is observability; away from the attractor they already differ at $O(1)$ before any observability argument, by an explicit, time-dependent, non-local map along the fluid worldline. Relative to Ref. [6], this note supplies the input a conformal-Fermi-coordinate treatment of a genuine non-attractor would need: the $O(k^0)$ divergence of the long mode’s comoving shift and the resulting displacement and dilation of the local frame. Relative to Refs. [7, 10], the agreement found there is explained as the $I \ll 1$ corner of the criterion rather than a coincidence of the USR transition. Relative to Refs. [8, 9], the coordinate-removal argument for the canonical single-field local f_{NL} is $O(\epsilon)$ -controlled in inflation; whether an analogous physical-coordinate argument removes the $O(1)$ in-in monopole in a matter-like contraction is a distinct, open question, not addressed here. Relative to Refs. [11, 12], the mechanism and order differ: their breakdown is localized to a finite window of scales at a slow-roll-to-USR transition and traced to a momentum mismatch, while the criterion here applies to all super-Hubble scales at $O(k^0)$, at constant ϵ , with no transition required.

V. LIMITS

The second-order map, Eqs. (3)–(5), is exact for constant ϵ and $c_s = 1$ (the scalar-field realization of a given w); it is not the second-order map for a genuine fluid with $c_s^2 = w \neq 1$, which needs the c_s -dependent second-order ADM constraints – an open, bounded extension. The USR second-order statement in Sec. III is structural (every frozen second-order kernel is proportional to ϵ , and vanishes as $\epsilon \rightarrow 0$), not a re-solve with a time-dependent $\epsilon(t) = \epsilon_s(a/a_s)^{-6}$; a full time-dependent- ϵ second-order calculation was not attempted. The linear criterion, Eq. (2), holds on super-Hubble scales for any single-field history and any c_s , from a flat, super-Hubble initial slice, up to the dropped $O(k_L^2/a^2 H^2)$ gradient term bounded in Sec. II; it is not literally exact at all k_L .

REPRODUCIBILITY STATEMENT

The linear criterion, second-order map, general- w closed forms, and all four validations are computed by exact sympy in the committed script research/theory_audit/separate-universe-failure-criterion.2026.09.04.py (SHA-256 21668ab6...), with results in the companion .json (SHA-256 4e2a49be...) and derivation in the companion note of the same base name (.md); the second-order threading map is derived in threading_map_second_order.2026.09.04.{md,py,json}, and the in-in inputs in fnl_monopole_adjudication.2026.09.03.md. The label-resolved second-order composition (both worldline labels, Eqs. (3)–(4)) and the restored gradient term (Eq. (2)) are

TABLE I. Validation of the criterion on four backgrounds (constant ϵ , $c_s = 1$ except USR). I is the ζ -growth-weighted mean of ϵ/c_s^2 ; $\lambda = 1 - I/3$; $f^{\text{in-in}}(\mu)$ the from-scratch in-in bispectrum and $f_{\text{mono}}^{\text{in-in}}$ its monopole; $f_{\delta N}^{\text{init}}$ the isotropic separate-universe value (initial-position label, the label the separate universe computes).

case	I	λ	$f^{\text{in-in}}(\mu)$	$f_{\text{mono}}^{\text{in-in}}$	$f_{\delta N}^{\text{init}}$	result
dust, $\epsilon = 3/2$	$3/2$	$1/2$	$-35/16 + \frac{15}{16}\mu^2$	$-15/8$	-5	$I = O(1)$: identification off by 25/8
USR, $\epsilon \propto a^{-6}$	$\ll 1$	$1 + O(\epsilon)$	$5/2$	$5/2$	$5/2$	$I = O(\epsilon)$: agree to $O(\epsilon)$
attractor slow roll	0	1	$\frac{5}{12}(1 - n_s)$	same	same	$I = 0$: identity map
ekpyrosis, dominant mode	0	1	attractor-like	same	same	$I = 0$: consistency check ($\dot{\zeta}_L = 0$ trivially)

derived and sympy-validated in `psu_gates.S1_S2.2026_09_04.{py,json}` (SHA-256 `4f641022... / f4164019...`). Figure 1 is generated by `make_fig_lambda_fmap.py` in this directory, transcribing the general- w closed forms with no new derivation. Manifest: `reproducibility/manifests/experiments/lift2-separate-universe-failure-criterion.json` and `reproducibility/manifests/experiments/psu-gates-s1-s2-label-composition-criterion.json`. Local CPU, \$0, under 5 seconds total.

AI USAGE DISCLOSURE

The scripts verify algebra: the symbolic manipulations underlying the linear criterion, the second-order map for both worldline labels, and the four validations were performed by deterministic sympy scripts written and executed under the author’s direction, with every intermediate result printed and cross-checked against the frozen inputs. The physical identification of the separate universe’s output with the initial-position-label δN_c (Sec. II), and the framing of the criterion as a change of variable rather than a failure (Sec. III), are the author’s arguments, not sympy output. Claude (Anthropic) assisted with literature cross-checks, prose drafting, and LaTeX formatting under the author’s direction; all physics content, equations, and numerical results were verified by the author against the committed script output.

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